24. Noncommutative Examples

1. $(2e+3a+0b)+(4e+2a+3b) = e+0a+3b$, where coefficients are added in $\mathbb{Z}_5$.

2. Because $\{e,a,b\}$ is a cyclic multiplicative group, we have $aa=b, bb=a, ab=ba=e$, and of course e acts as identity element. With the coefficients of e,a, and b from $\mathbb{Z}_5$, we obtain

$$\begin{gathered}(2e+3a+0b)(4e+2a+3b) = \\ 2e(4e+2a+3b)+3a(4e+2a+3b)+0b(4e+2a+3b) = \\ (3e+4a+1b)+(4e+2a+1b)+(0e+0a+0b) = 2e+a+2b.\end{gathered}$$

3. Because $\{e,a,b\}$ is a cyclic multiplicative group, we have $aa=b, bb=a, ab=ba=e$, and of course e acts as identity element. With the coefficients of e,a, and b from $\mathbb{Z}_5$, we obtain

$$\begin{aligned}(3e+3a+3b)^2 &= 3e(3e+3a+3b)+3a(3e+3a+3b)+3b(3e+3a+3b) \\ &= (4e+4a+4b)+(4e+4a+4b)+(4e+4a+4b) = 2e+2a+2b.\end{aligned}$$

Having seen how this could have been simplified in view of the equal coefficients of e,a, and b, we can now proceed more quickly and find that $(3e+3a+3b)^4 = (2e+2a+2b)^2 = 4\cdot(1e+1a+1b)^2 = 4\cdot(3e+3a+3b) = 2e+2a+2b$.

4. $(i+3j)(4+2j-k) = 4i+2ij-ik+12j+6jj-3jk = 4i+2k+j+12j-6-3i = -6+i+13j+2k$.

5. $i^2j^3kji^5 = (-1)(-j)kji = (jk)(ji) = i(-k) = j$.

6. $(i+j)^{-1} = \frac{1}{i+j}\cdot\frac{-i-j}{-i-j} = \frac{-i-j}{2} = -\frac{1}{2}i-\frac{1}{2}j$.

7. $[(1+3i)(4j+3k)]^{-1} = (4j+3k+12k-9j)^{-1} = (-5j+15k)^{-1} = \frac{1}{-5j+15k}\cdot\frac{5j-15k}{5j-15k} = \frac{5j-15k}{25+225} = \frac{j-3k}{50} = \frac{1}{50}j-\frac{3}{50}k$.

8. $(0\rho_0+1\rho_1+0\rho_2+0\mu_1+1\mu_2+1\mu_3)(1\rho_0+1\rho_1+0\rho_2+1\mu_1+0\mu_2+1\mu_3)$
$= (1\rho_1+1\rho_2+1\mu_3+1\mu_2)+(1\mu_2+1\mu_3+1\rho_2+1\rho_1)+(1\mu_3+1\mu_1+1\rho_1+1\rho_0)$
$= (1\rho_0+1\rho_1+0\rho_2+1\mu_1+0\mu_2+1\mu_3)$

9. The center is $\{r+0i+0j+0k \mid r\in R, r\neq 0\}$ because nonzero coefficients of i,j, or k lead to an element that does not commute with j,k, or i respectively.

10. Clearly $\{a+bi \mid a,b\in\mathbb{R}\}$ and $\{a+bj \mid a,b\in\mathbb{R}\}$ as well as $\{a+bk \mid a,b\in\mathbb{R}\}$ are subrings of the quaternions that are actually fields isomorphic to $\mathbb{C}$.

11. F F F F F F T F T F

12. **a.** The polynomial $x^2+1\in\mathbb{H}[x]$ has $i,-i,j,-j,k$, and $-k$ as zeros in $\mathbb{H}$.

b. The subset $\{1,-1,i,-i,j,-j,k,-k,\}$ of $\mathbb{H}$ is a group under quaternion multiplication and is not cyclic because no element has order greater than 2.

13. Let $\psi(m,n) = (m,-m)$. It is easily seen that ψ is an endomorphism of $\mathbb{Z}$. Then $(\phi\psi)(m,n) = \phi(\psi(m,n)) = \phi(m,-m) = (0,0)$.

14. The matrices

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix},$$

have inverses

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix}, \text{ and } \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}, \text{ respectively in every}$$

field. (Note that if the field has characteristic 2, we have -1 = 1 so the last four matrices in the second row may be the same as four in the top row.)

15. Let $m \in \mathbb{Z}$ [or $m \in \mathbb{Z}_n$ as the case may be]. Let ϕ_m be the endomorphism of the additive abelian group of the ring such that $\phi_m(1) = m$. Then $\{\phi_m \mid m \in \mathbb{Z}$ [or $m \in \mathbb{Z}_n]\}$ is the entire homomorphism ring, because a homomorphism of each of these cyclic groups is determined by its value on the generator 1 of the group. Define $\psi : \text{End}(\mathbb{Z}) \to \mathbb{Z}$ [or $\text{End}(\mathbb{Z}_n) \to \mathbb{Z}_n$] by $\psi(\phi_m) = m$. Now $(\phi_i + \phi_j)(1) = \phi_i(1)+\phi_j(1) = i+j = \phi_{i+j}(1)$, so $\phi_i+\phi_j = \phi_{i+j}$ because these homormophisms agree on the generator 1. Hence $\psi(\phi_i + \phi_j) = \psi(\phi_{i+j}) = i + j = \psi(\phi_i) + \psi(\phi_j)$, so ψ is an additive homomorphism. Also, $(\phi_i\phi_j)(1) = \phi_i(\phi_j(1)) = \phi_i(j) = ij = \phi_{ij}(1)$, so $\phi_i\phi_j = \phi_{ij}$. Therefore $\psi(\phi_i\phi_j) = \psi(\phi_{ij}) = ij = \psi(\phi_i)\psi(\phi_j)$. Hence ψ is a ring homomorphism. By definition, the image under ψ is the entire ring $\mathbb{Z}$ [or $\mathbb{Z}_n$]. If $\psi(\phi_i) = \psi(\phi_j)$, then $i = j$ in $\mathbb{Z}$ [or $\mathbb{Z}_n$] so ϕ_i and ϕ_j map the generator 1 into the same element and thus are the same homomorphism. Thus ψ is an isomorphism.

16. A homomorphism of $\mathbb{Z}_2 \times \mathbb{Z}_2$ can map the generators (1, 0) and (0, 1) onto any elements of the ring. Thus there are a total of $4 \cdot 4 = 16$ homomorphisms of $\mathbb{Z}_2 \times \mathbb{Z}_2$ into itself, while the ring itself has only 4 elements . Thus the ring of all homomorphisms cannot be isomorphic to the ring itself, for they have different cardinality.

17. Because we are dealing with homomorphisms, it suffices to show that $(YX - XY)(ax^n) = 1(ax^n)$ for a monomial $ax^n \in F[x]$. We have

$$\begin{aligned} (YX - XY)(ax^n) &= (YX)(ax^n) - (XY)(ax^n) \\ &= Y(ax^{n+1}) - X(nax^{n-1}) \\ &= (n+1)ax^n - nax^n = 1(ax^n). \end{aligned}$$

18. Let $\phi : RG \to R$ be defined by $\phi(re) = r$. Then $\phi(re + se) = \phi((r + s)e) = r + s = \phi(re) + \phi(se)$, and $\phi((re)(se)) = \phi((rs)e) = rs = \phi(re)\phi(se)$, so ϕ is a homomorphism. Clearly, the image of RG under ϕ is all of R. If $\phi(re) = \phi(se)$, then $r = s$, so ϕ is one to one. Thus ϕ is an isomorphism.

19. a. From the statement of the problem, we expect the identity matrix I_2 to play the roll of 1, the matrix B with coefficient b to play the role i, and the matrix C with coefficient c to play the roll of j in a quaternion $a + bi + cj + dk$. Note that $B^2 = C^2 = -I_2$. Thus we let

$$K = BC = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix} = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}.$$

b. (See the answer in the text.)

c. We should check that ϕ is one to one.